

An Introduction to 3D Printing

Part 2: Practical Application

Examples, Critical Thinking, and Design

Outline

Part 1: Printers

Process

Limitations

Dimensional Accuracy

Advantages

Part 2: Practical Application

Examples

Critical Thinking

Design

Part 3: Science

Vibrations and Waves

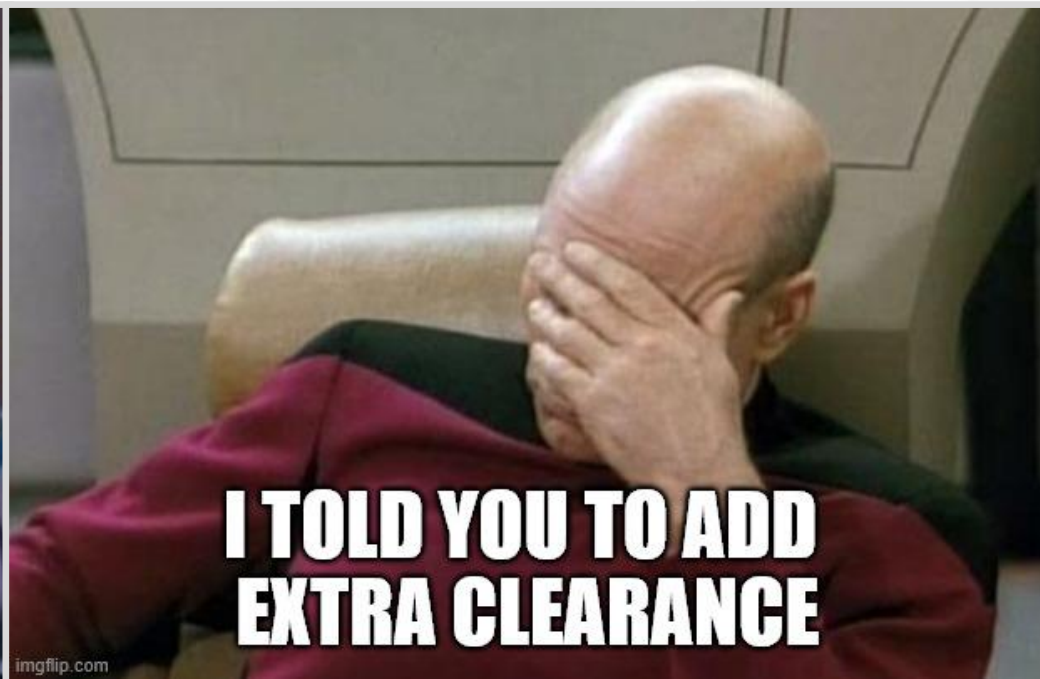
Linear Motion

Thermal Expansion

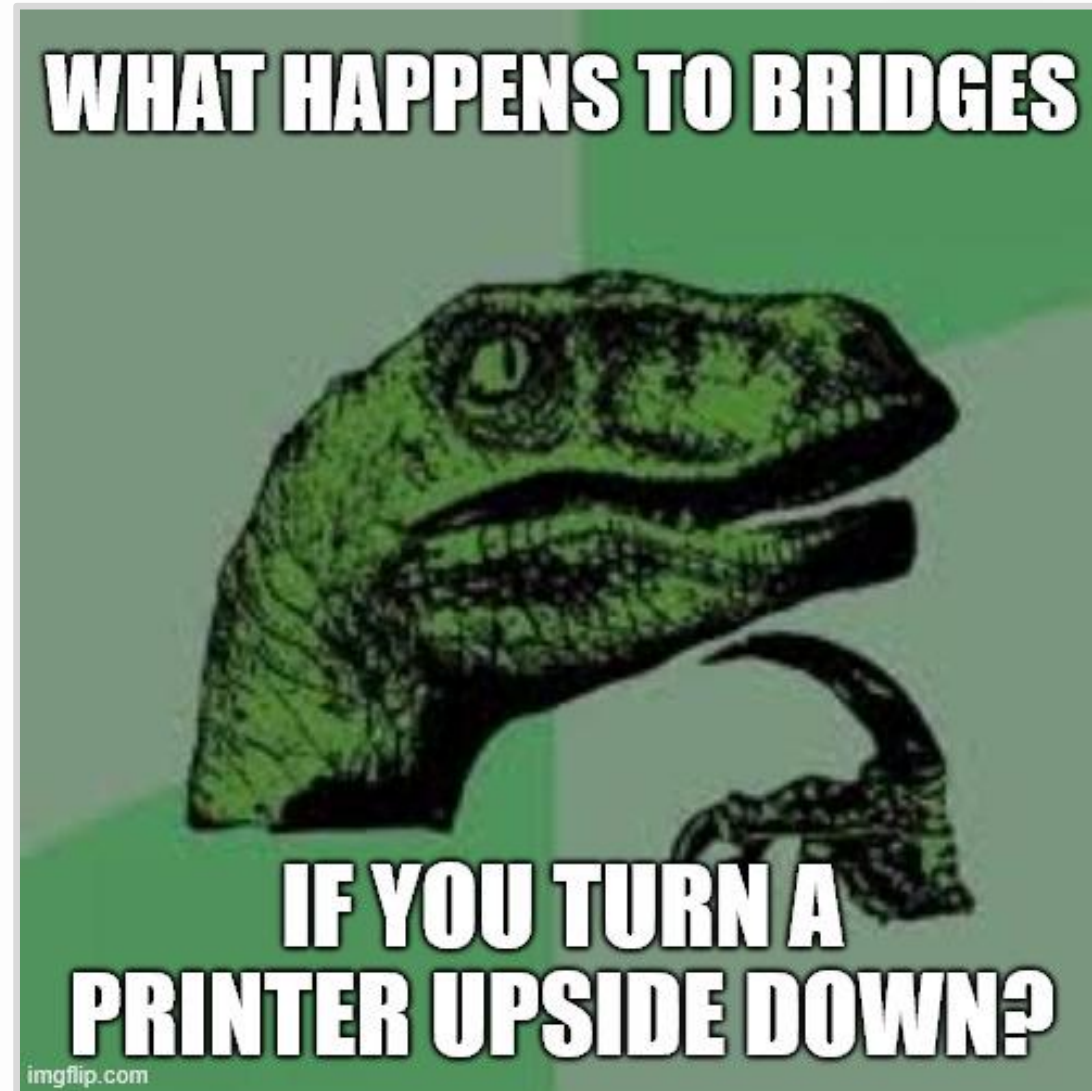
Pop Quiz

How do you help printed parts fit together?

- You sometimes need scale factor.
- You **always** need constant offset.



Anti-Gravity?



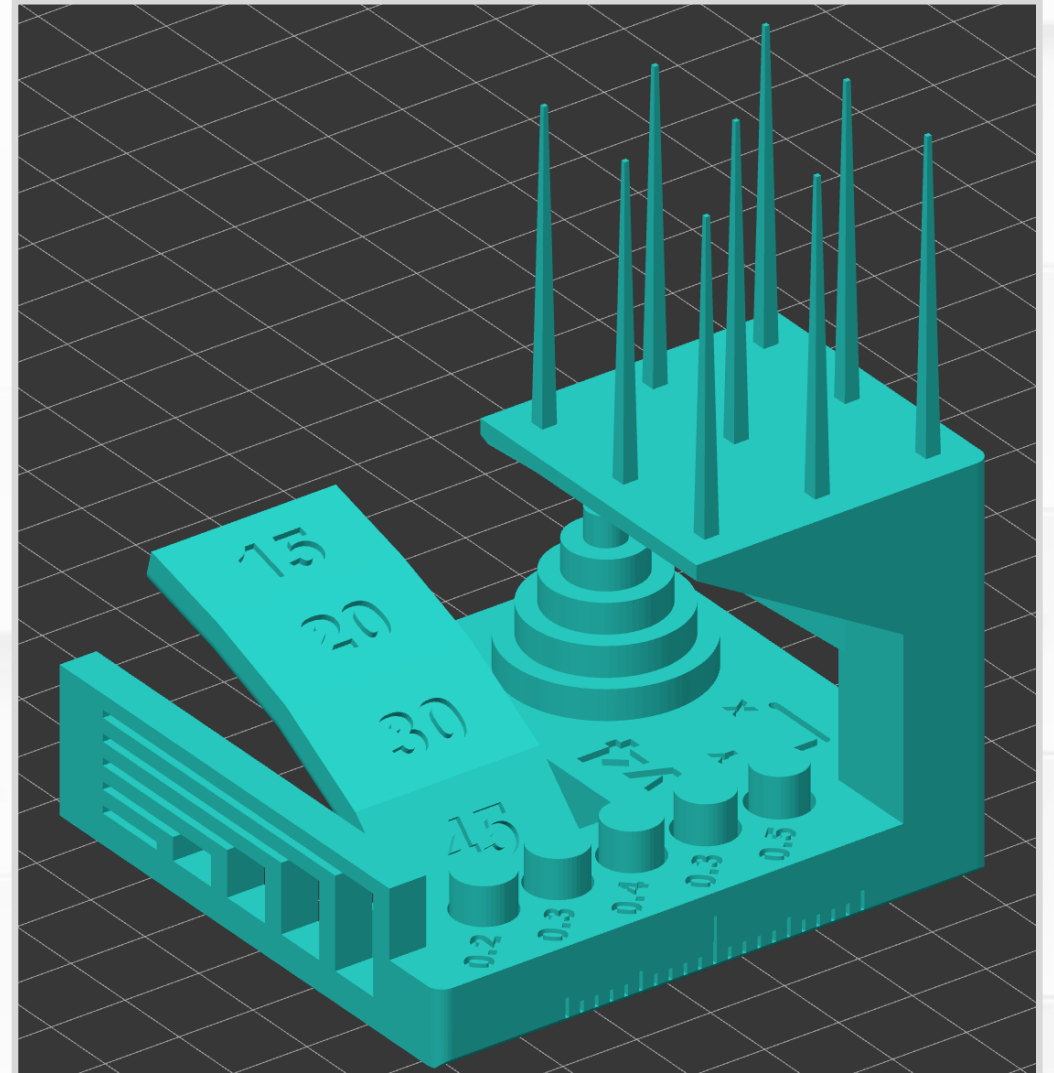
Anti-Gravity

- Positron GitHub.
 - Prints upside down.
 - Folds up and fits in a small box.
- Design team and users report the same print quality. Why?
- Extrusions hold their shape when you press them onto a solid surface.



Stress Testing

- Autodesk Test Print. One of many test prints.
- Difficult shapes that highlight **capabilities** and **limitations** of your printer + filament.
- What are these shapes testing?
- Full breakdown [here](#).
- Bridges, overhangs, print-in-place tolerance, dimensional accuracy, retractions, vibrations, ...



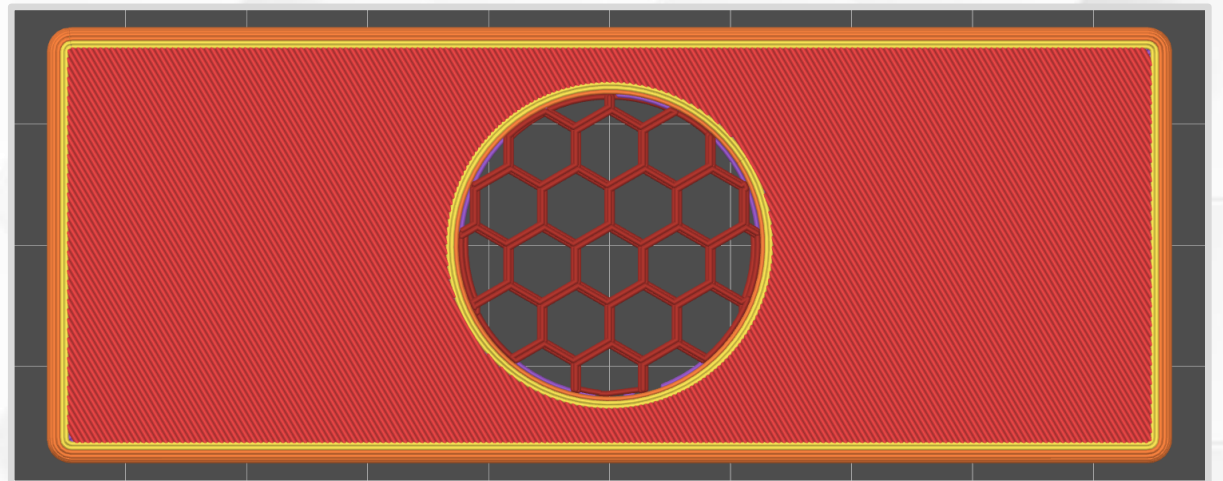
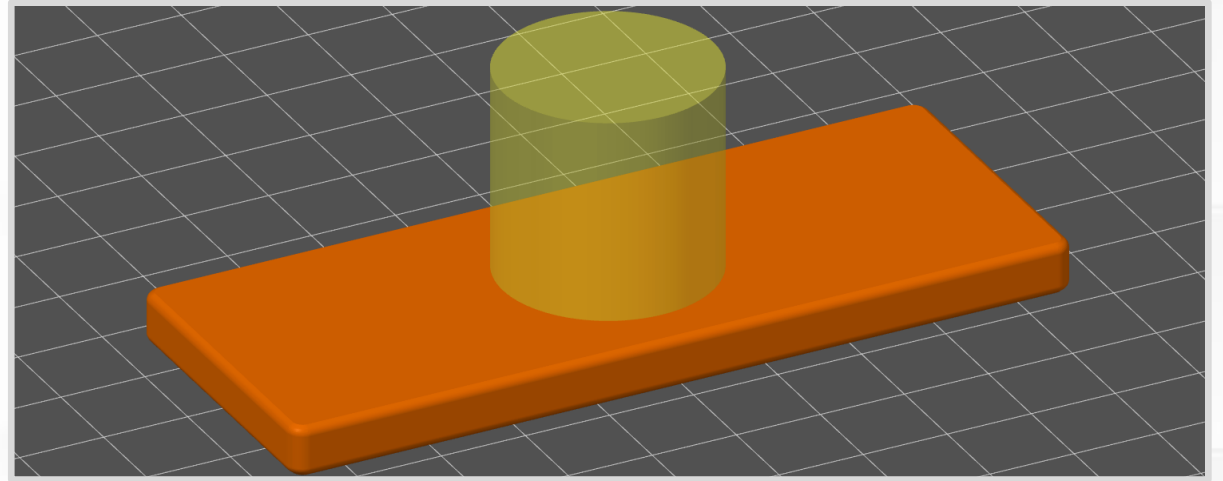
Corners

- Why is rounding corners with fillets a good idea?
- Stresses concentrate in corners.
 - Adding fillets improves strength.
 - Fillets on bottom surfaces reduce first layer warping.
- Print faster with more accurate movements. Why?
 - Toolhead is like a racecar going around a track.
 - Smoothing changes of direction means:
Less slowdown.
Less vibrations.

Advanced Slicing: Modifiers

Let's make a mesh grill with PrusaSlicer.

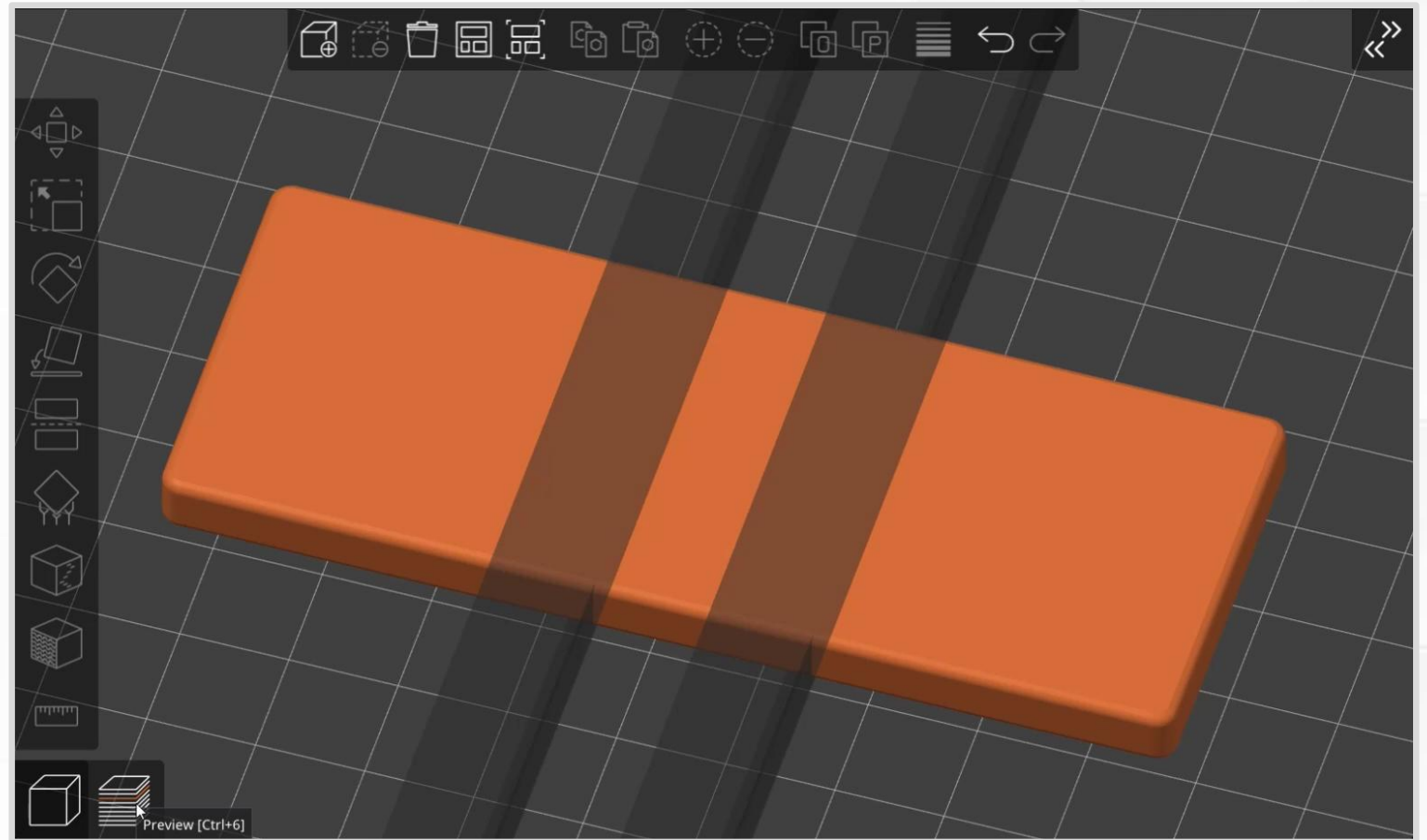
- Load STL file (rectangle).
- Add basic shape (cylinder) and change it to a modifier.
- Assign zero top and bottom layers.
- Set an infill pattern.
 - Hexagon makes a nice mesh.
- Change infill percent to alter the mesh size.



Advanced Slicing: Negative Volume

Now let's make a print-in-place hinge with PrusaSlicer.

- Start with a flat object.
- Download [Flexify template](#) that matches part height.
- Templates define negative space for PIP hinges.

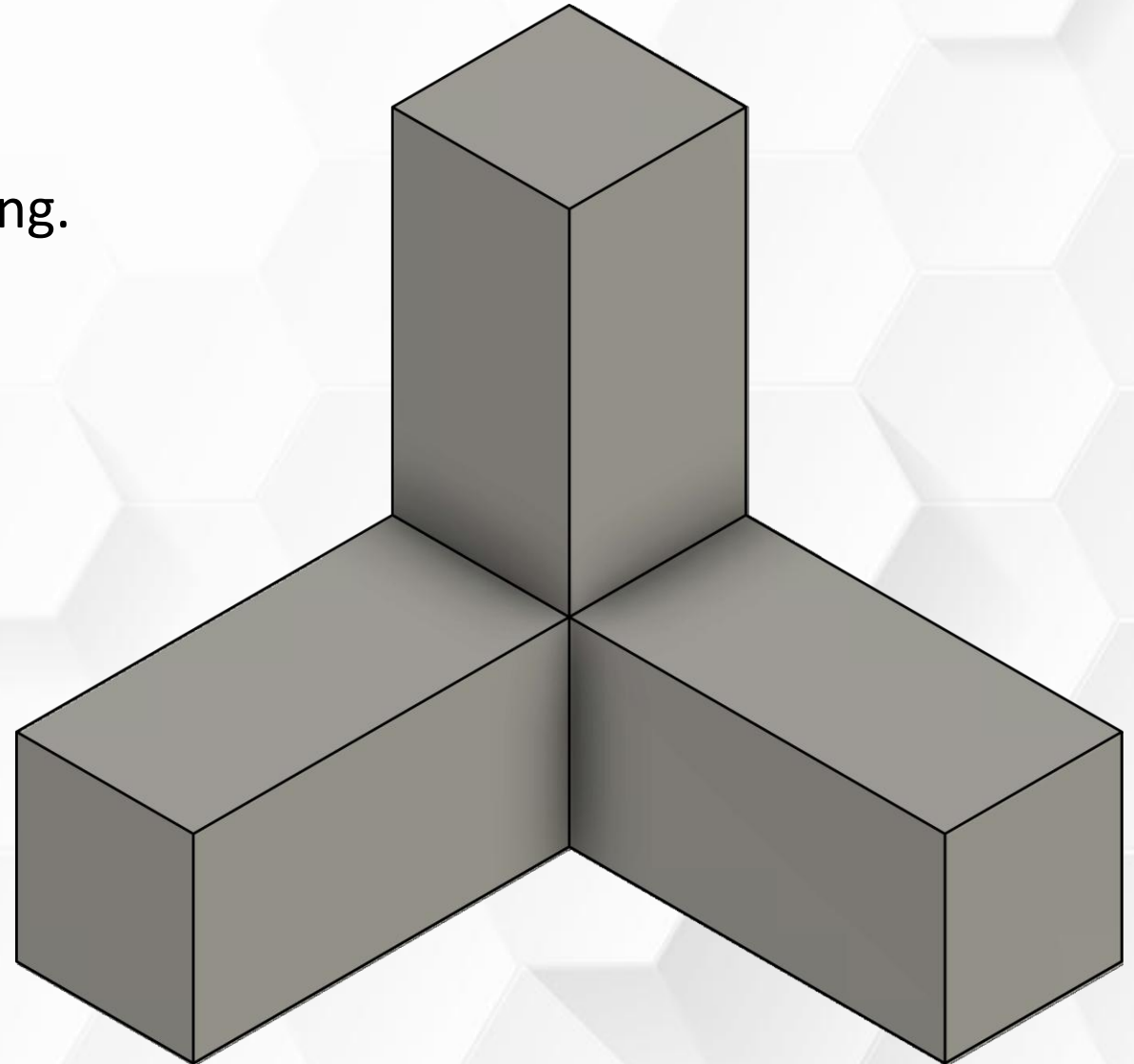


(Animation, [FlyingGyroscope GitHub](#))

Corner Bracket

Let's improve this 3-way corner bracket.

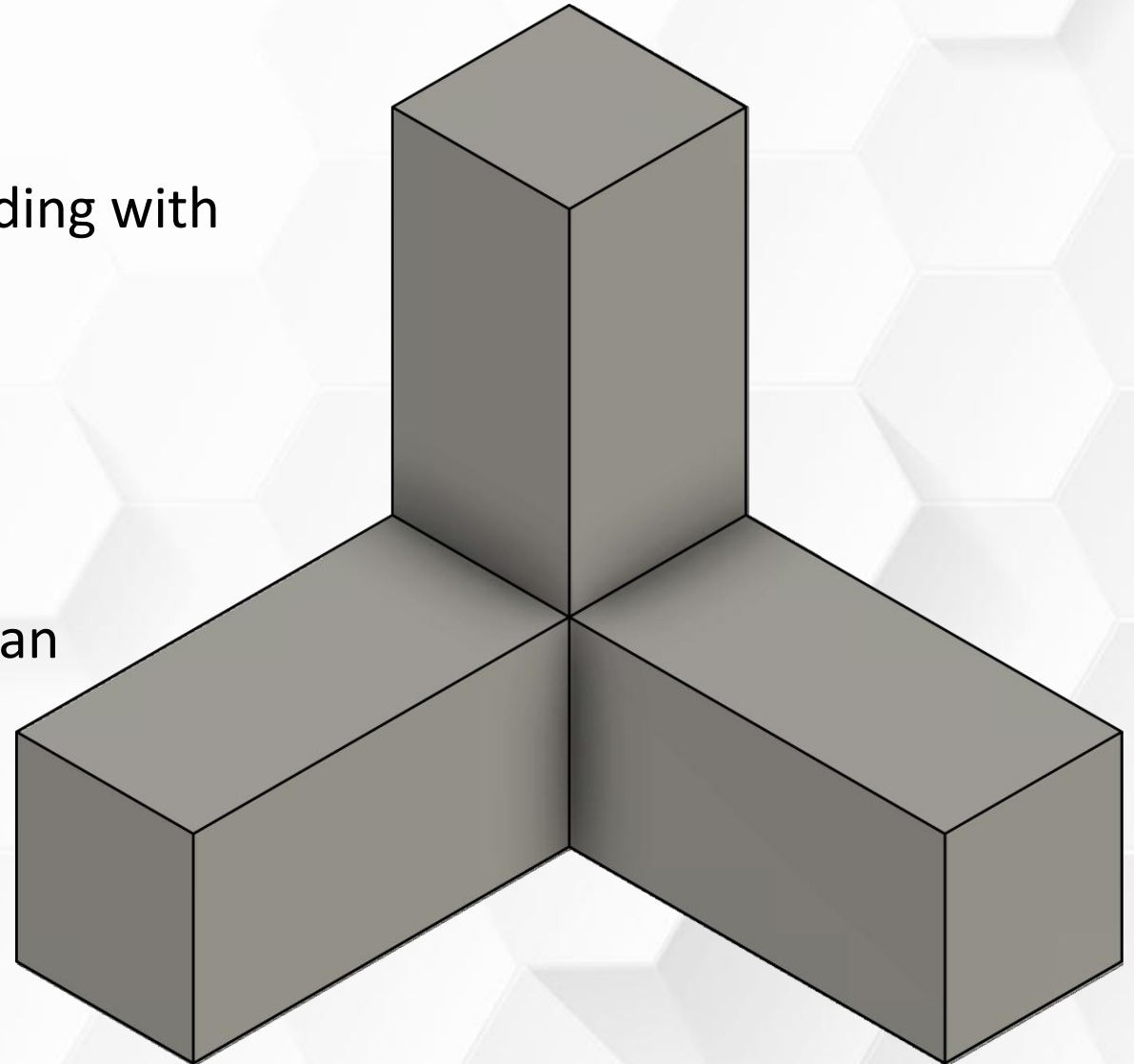
- Top section is weaker because of layer stacking.
- How can we make it stronger?
 - Change slicer settings?
 - Modify or reimagine the design?



Slicer Settings

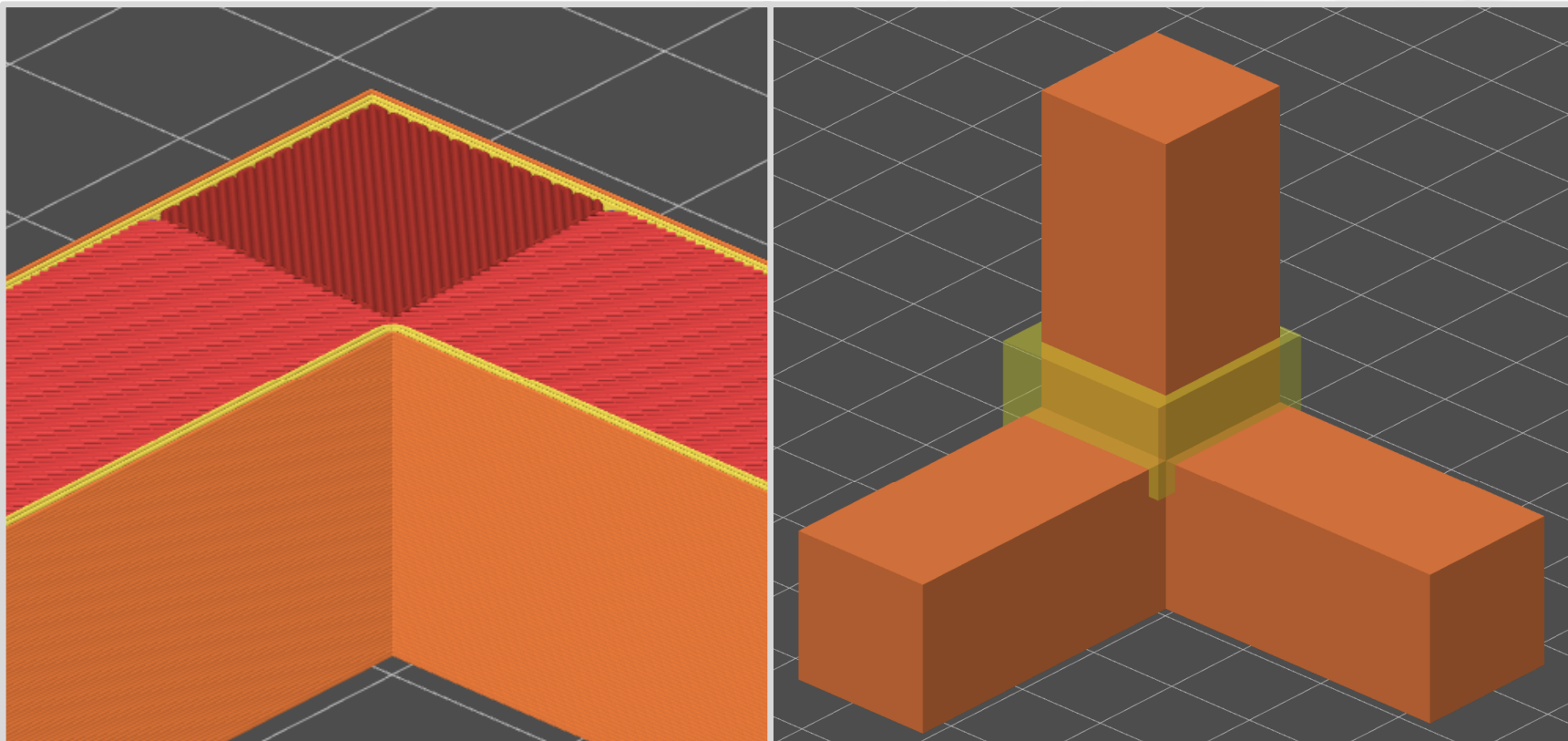
Let's start with slicer settings.

- How can we improve strength and layer bonding with slicer settings?
- Higher temperature.
- Slower printing speed.
- Less cooling.
- Increasing perimeters adds more strength than increasing infill density.
- Solid infill.
- Intentionally over-extrude by a small amount



Slicer Settings

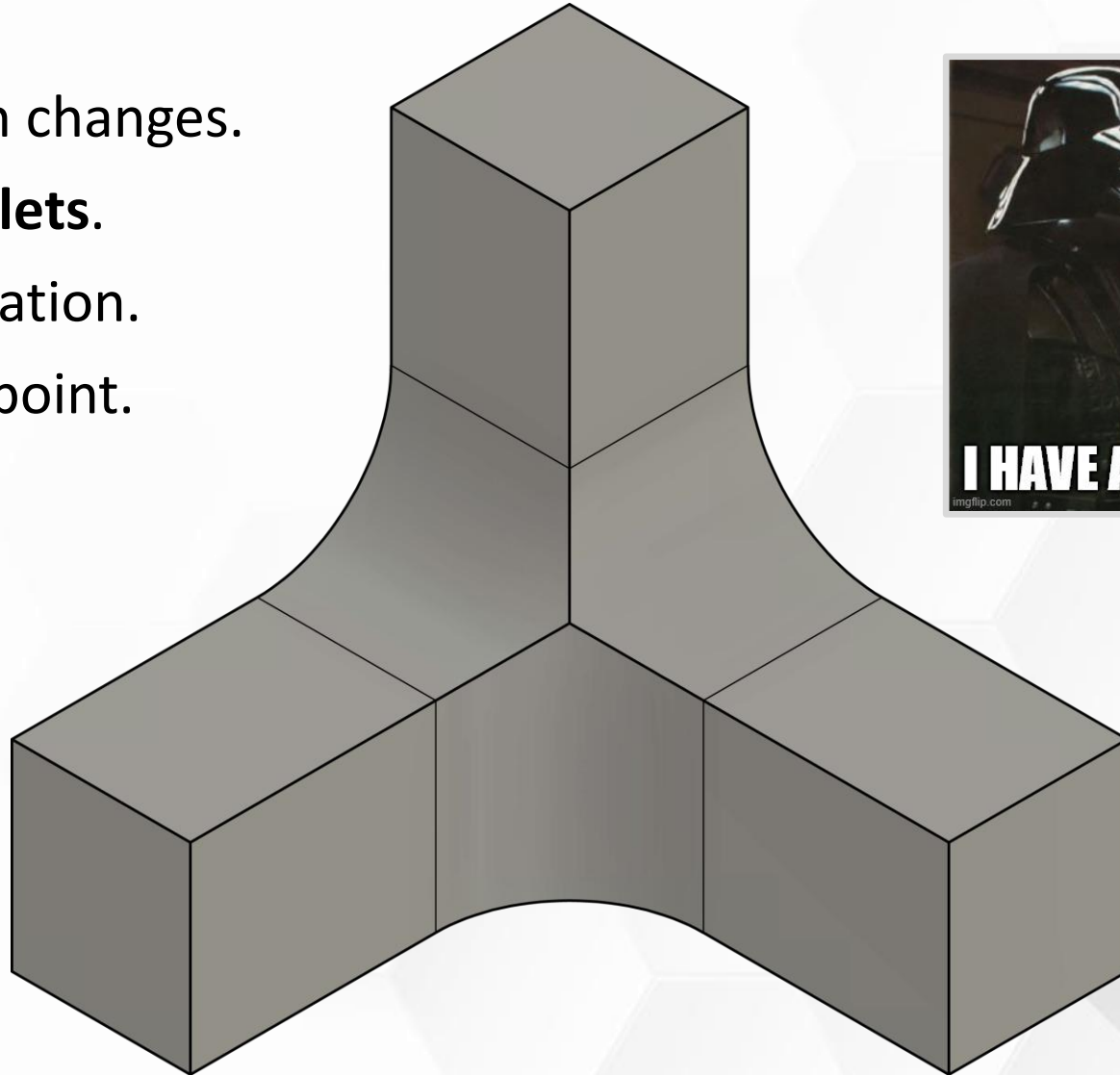
- Use modifier on just the weakest spot. (Yellow box on the right.)
- Assign 100% infill. Maximum surface area for layer bonding.



Corner Bracket Remix

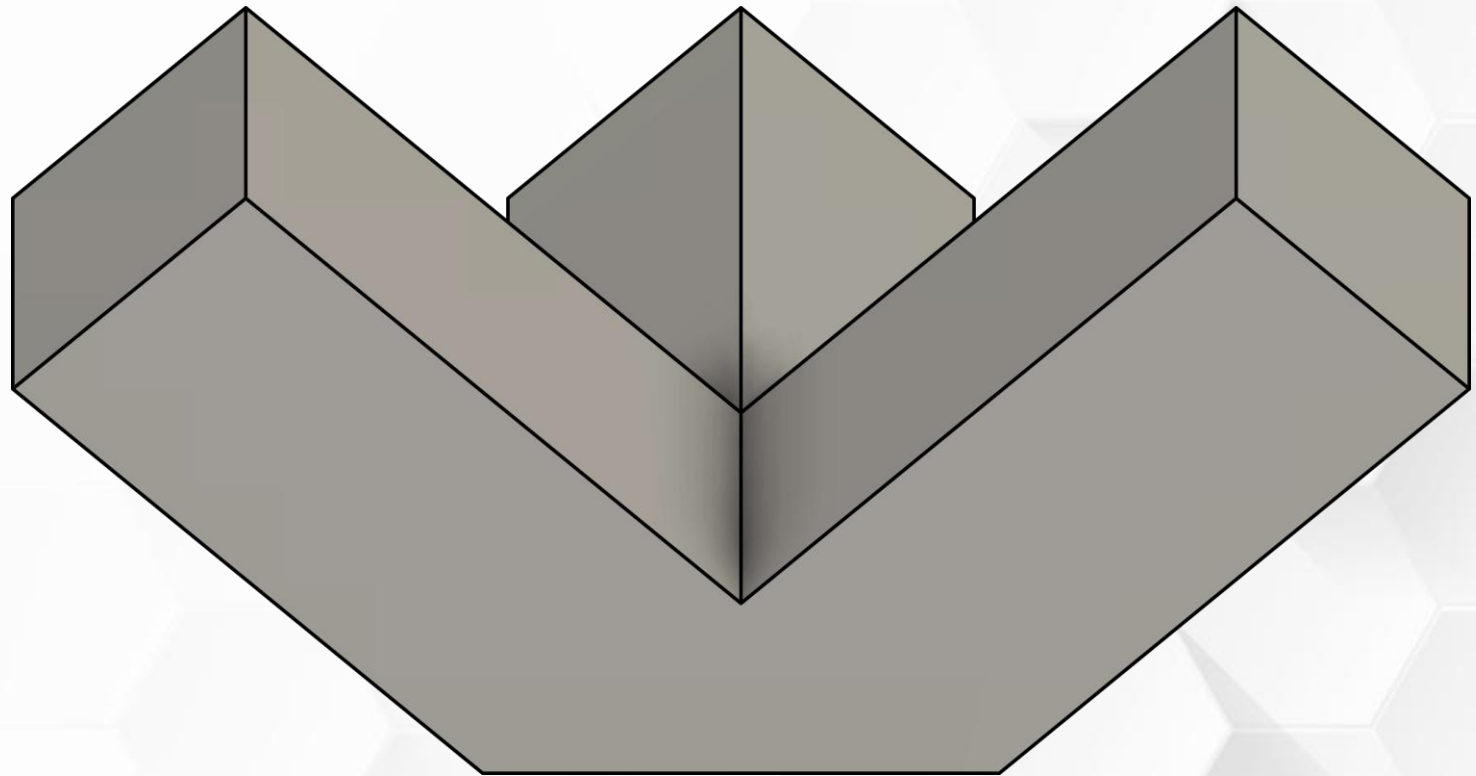
Now let's consider design changes.

- Reinforce joints with **fillets**.
- Reduce stress concentration.
- Reinforce the weakest point.
- Simple and effective.



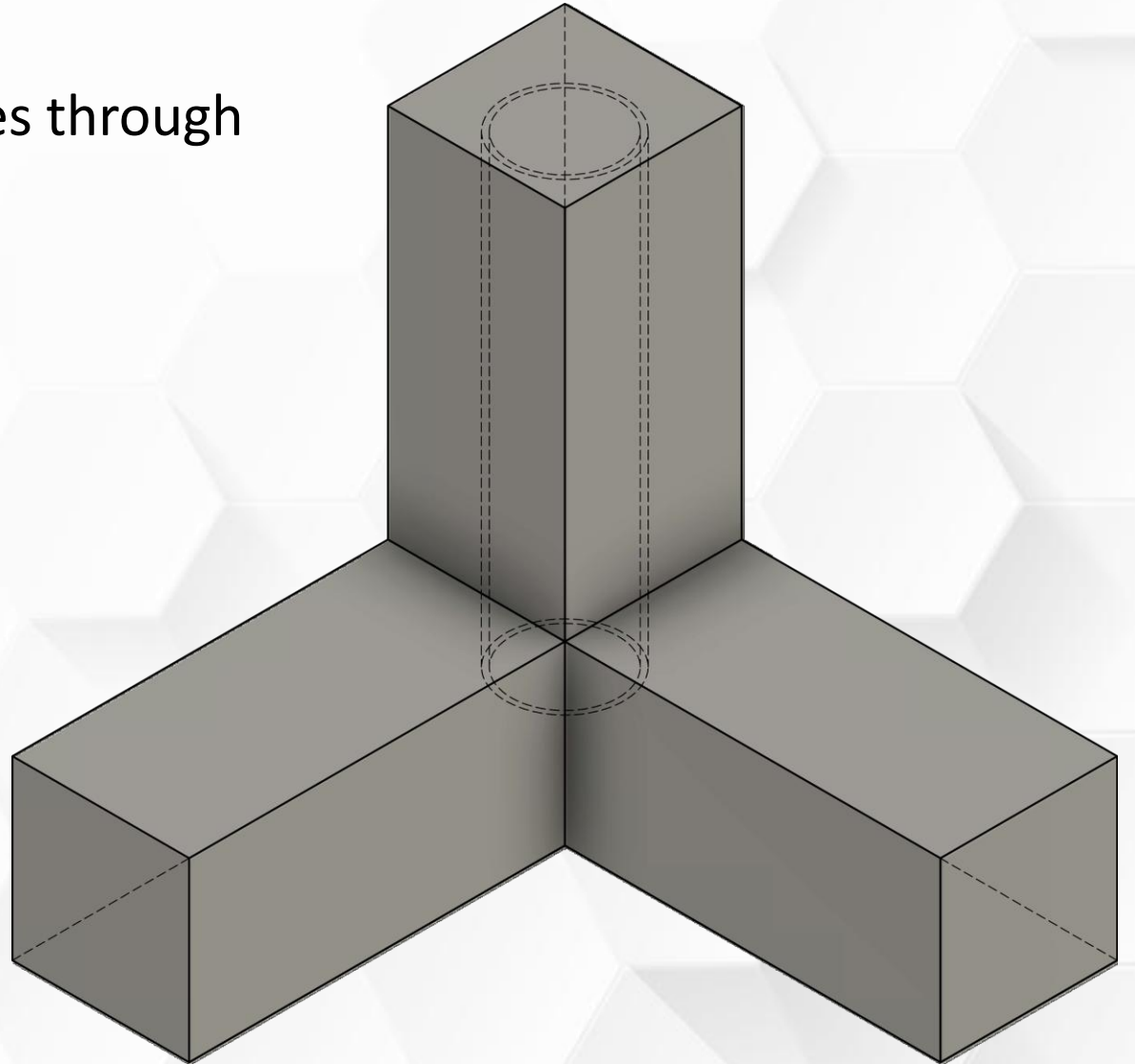
Corner Bracket Remix

- Angled print orientation.
- Trim the corner.
- Adds overhangs.
- Smaller first layer.
- Each section has equal strength.



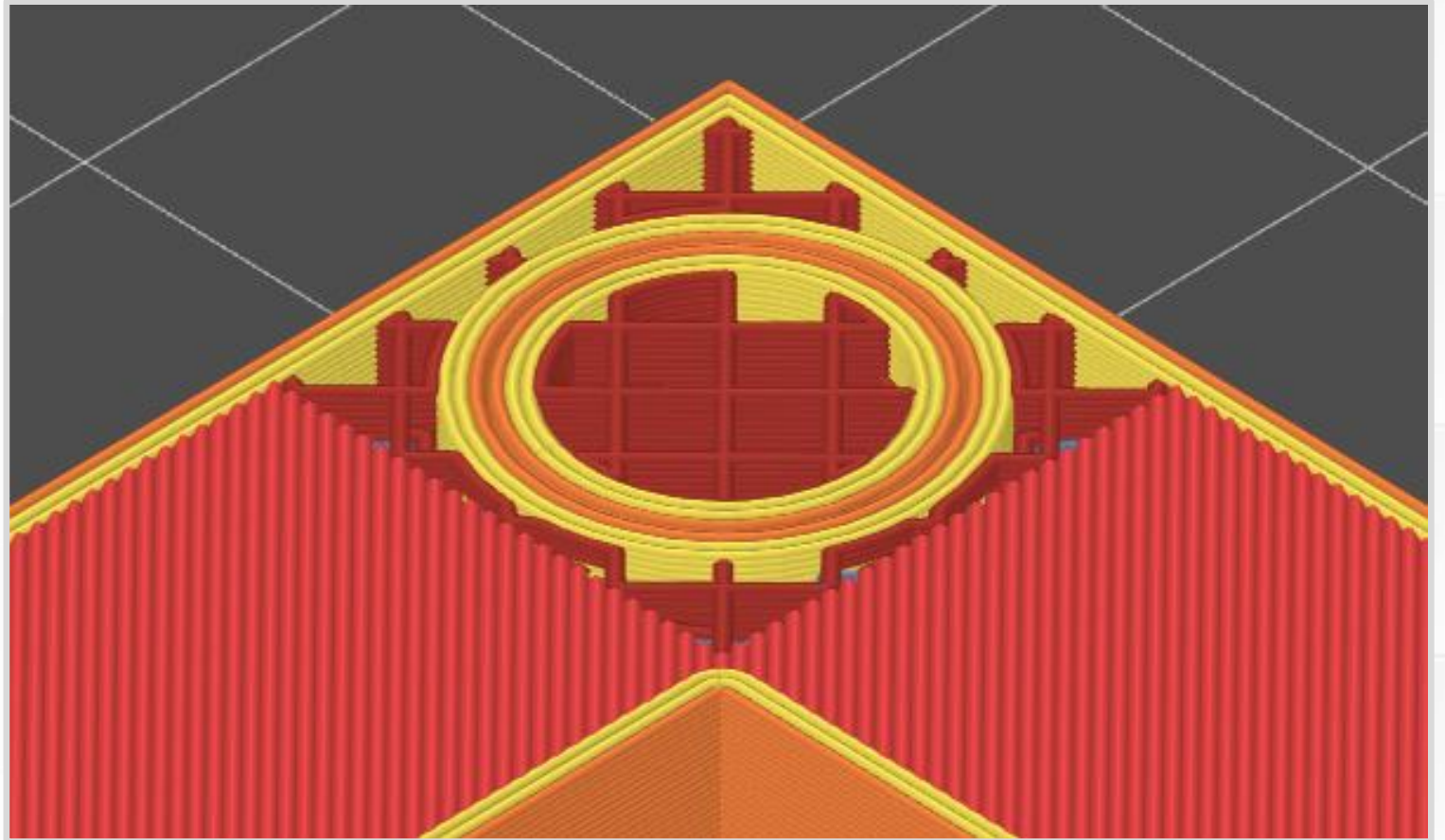
Corner Bracket Remix

- Cut a small slit (around 0.1 mm) that goes through upper and lower sections.
- Why?
- *Hint:* What will the slicer do?



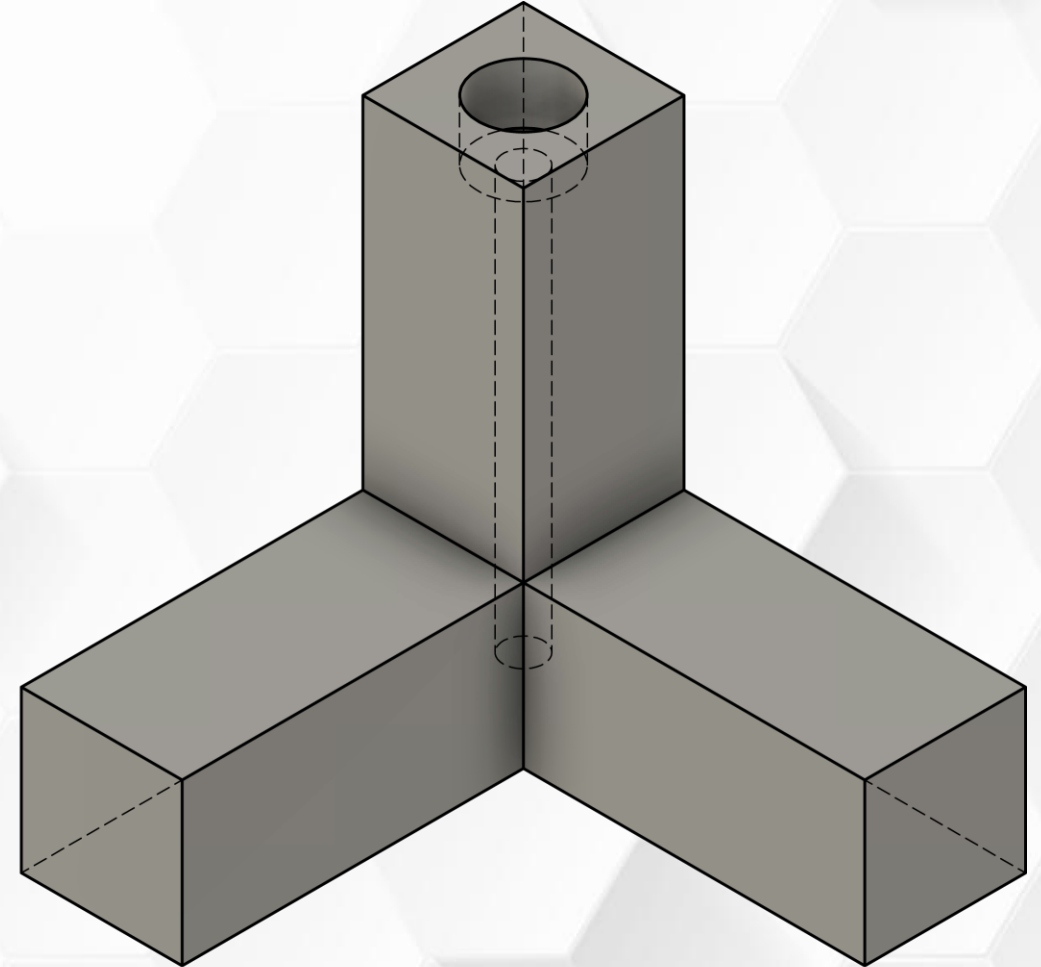
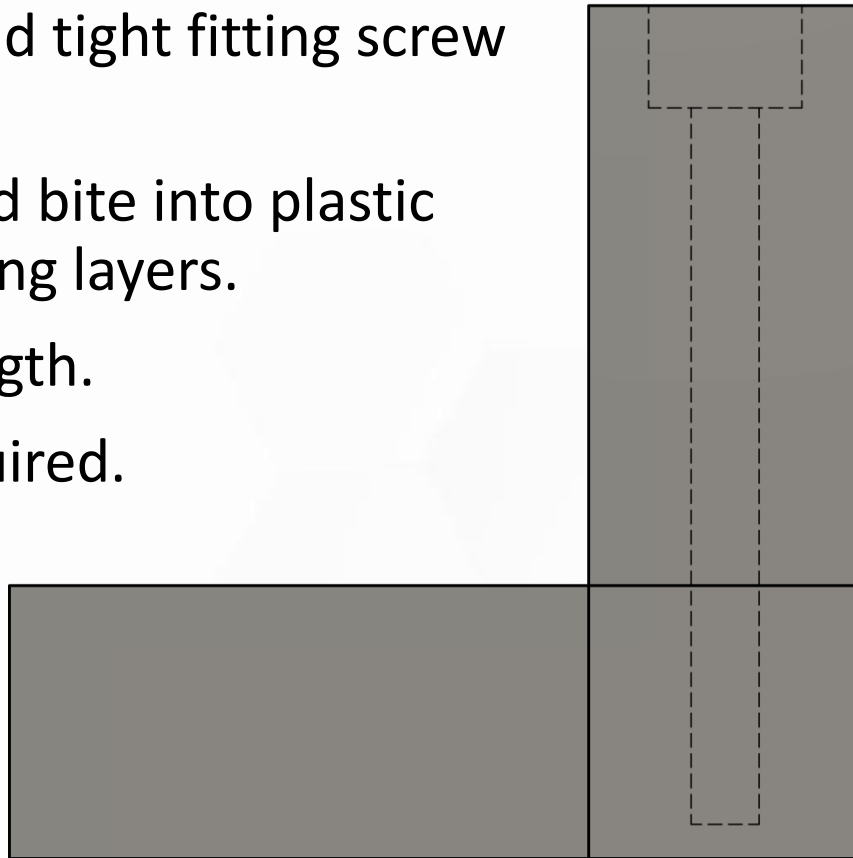
Corner Bracket Remix

- Tricks slicer into adding extra perimeters.
- More plastic and more area for layer bonding.
- Less effective than 100% infill modifier from earlier.



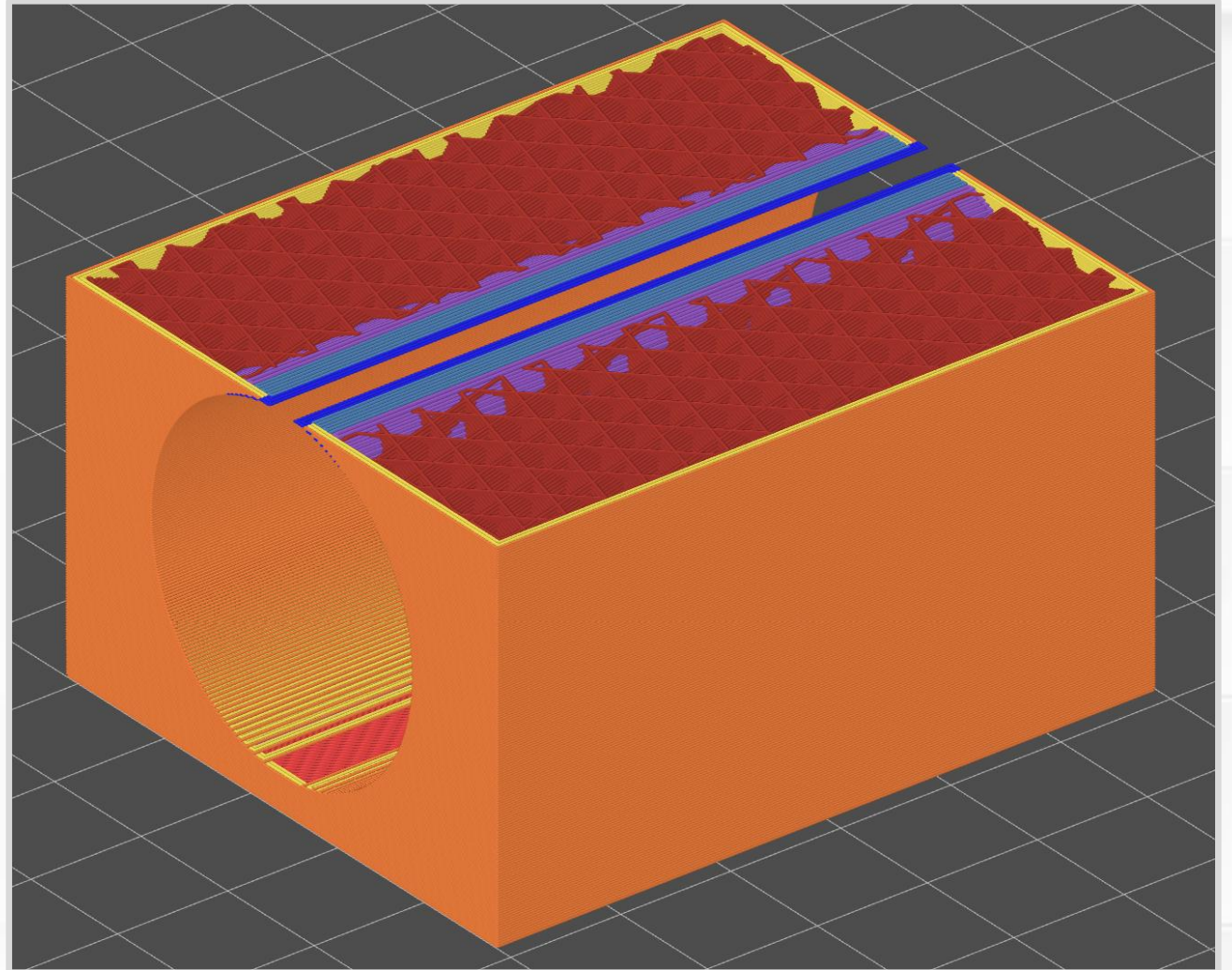
Corner Bracket Remix

- Borrow strength from something else.
- Cut a hole. Add tight fitting screw after printing.
- Threads should bite into plastic without cracking layers.
- Superior strength.
- Assembly required.



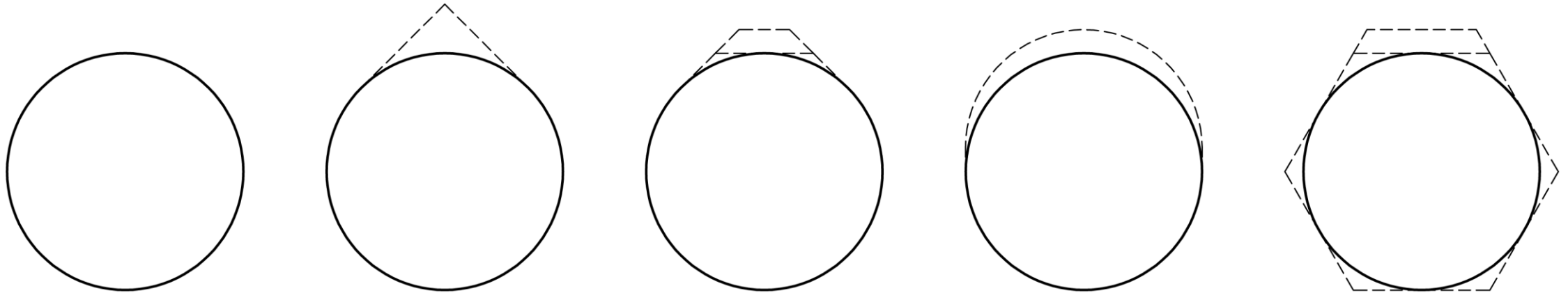
Horizontal Holes

- Horizontal holes have steep overhangs at the top.
- Hole is undersized if the extrusions collapse.
- Assume the hole must remain horizontal.
- How can you alter the design to fix the problem?



Horizontal Holes

- Make the top profile more printer-friendly.
- Add clearance at the top.



Bridges

- What can you do to improve bridges? (Besides supports.)
- *Hint:* How can you help extrusions lock in place?
 - Lower filament temperature.
 - Slower printing speed.
 - Increase cooling.
- Reimagine the object as multiple pieces.
 - Each piece is free from supports and bridges.
 - Assembly and clearances required.
- Can you reimagine the bridge?

Sequential Bridges

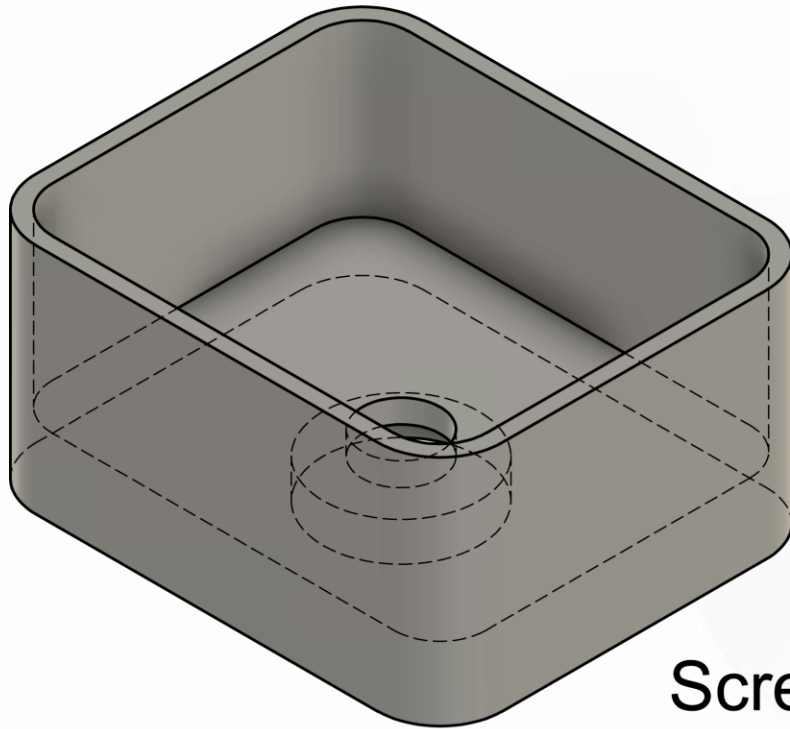
Let's combine small bridges to create a larger, multi-step bridge.



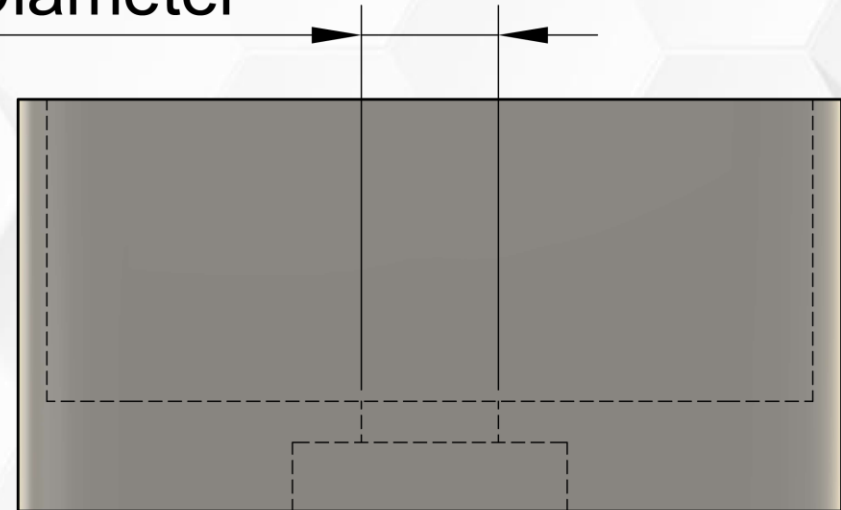
Sequential Bridges

Consider a storage box with a countersunk screw on the bottom.

- Can you see the hidden problem?



Screw Diameter



Screw Head Diameter

WARNING:

Detected print stability issues:

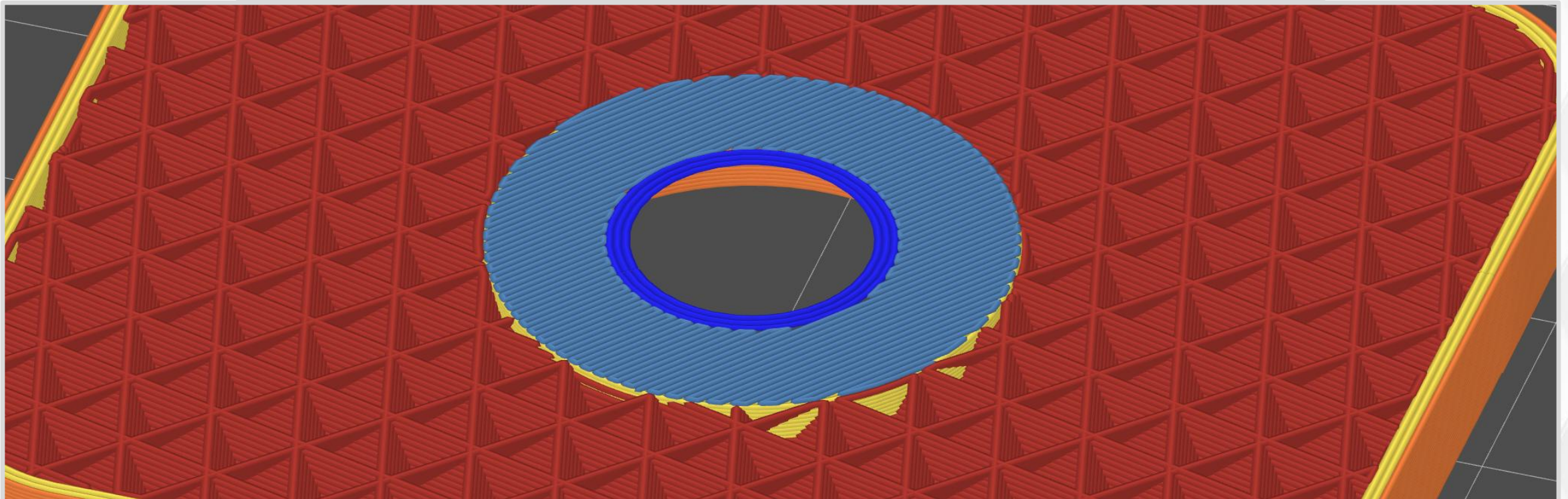


Box Bridge1

Floating bridge anchors, Long bridging extrusions

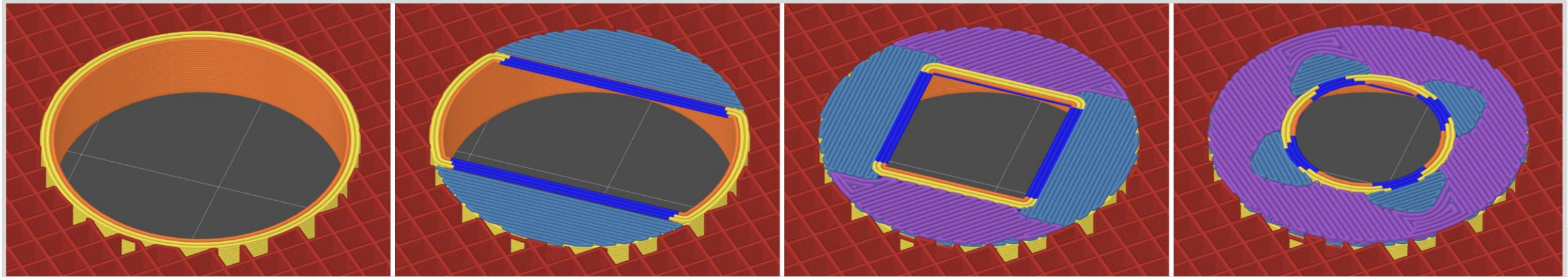
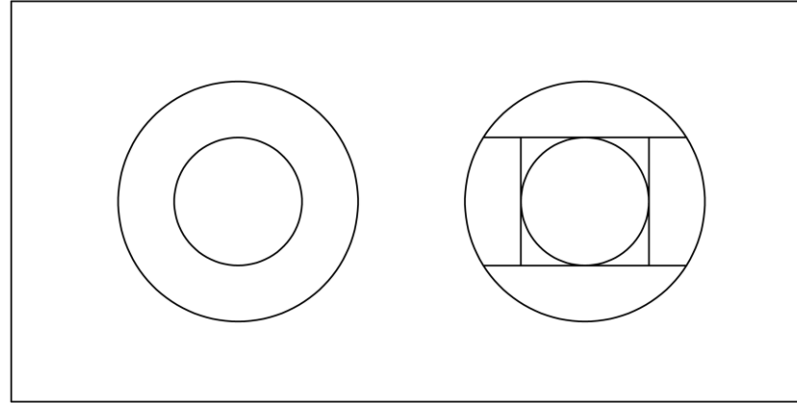


Consider enabling supports.



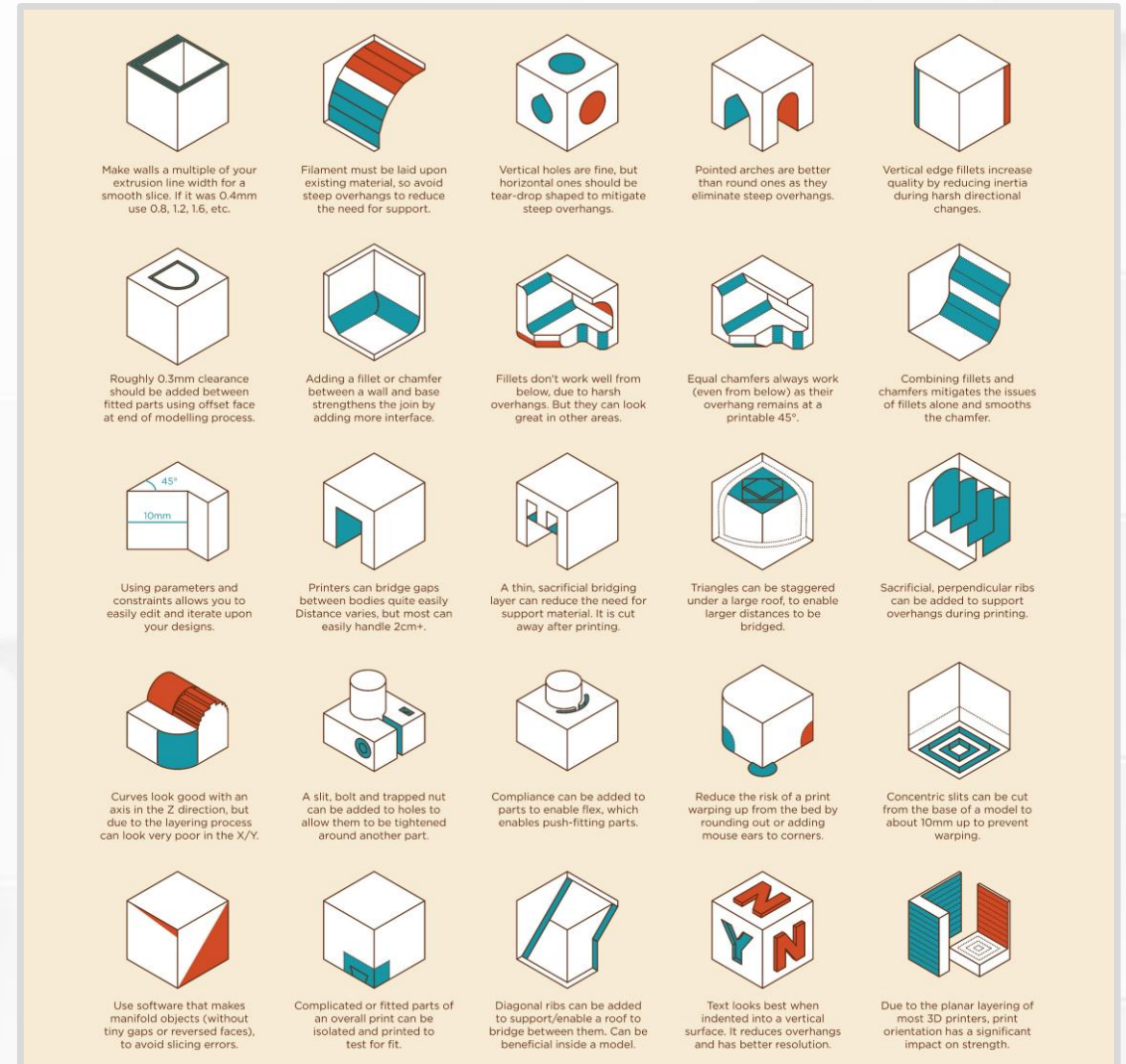
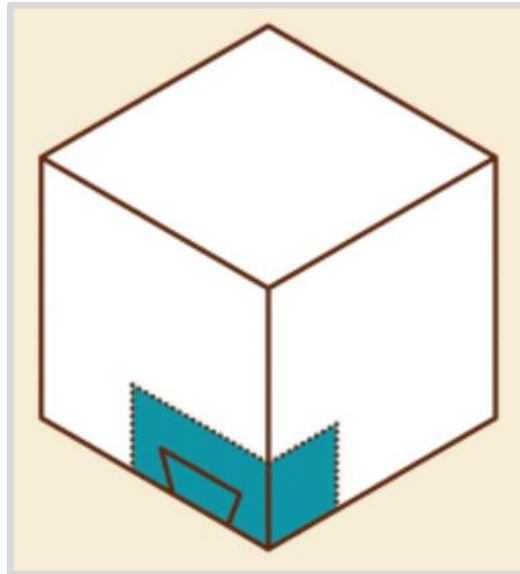
Sequential Bridges

- There are floating extrusions!
- We can fix this without supports.
- Use right-most drawing to stack three bridges.



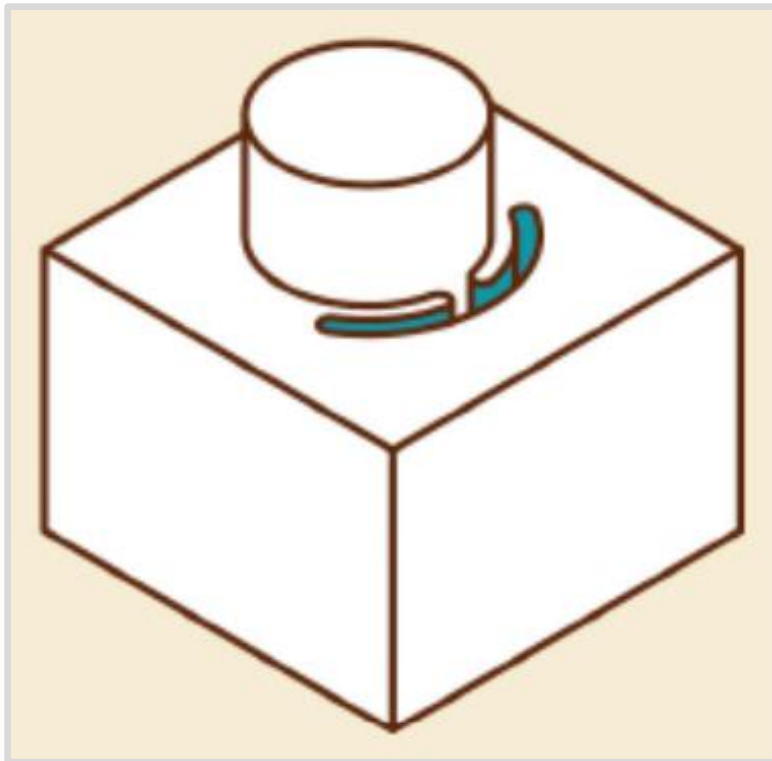
More Design Tips

- CAD Design Tips for 3D Printing
- Check out this great poster by Billie Ruben.
- “Complicated or fitted parts of a print can be isolated and printed to test for fit.”



Even More Design Tips

- “Compliance can be added to parts to enable flex, which enables push-fitting parts.”
- “Text looks best when indented into a vertical surface. It reduces overhangs and has a better resolution.”



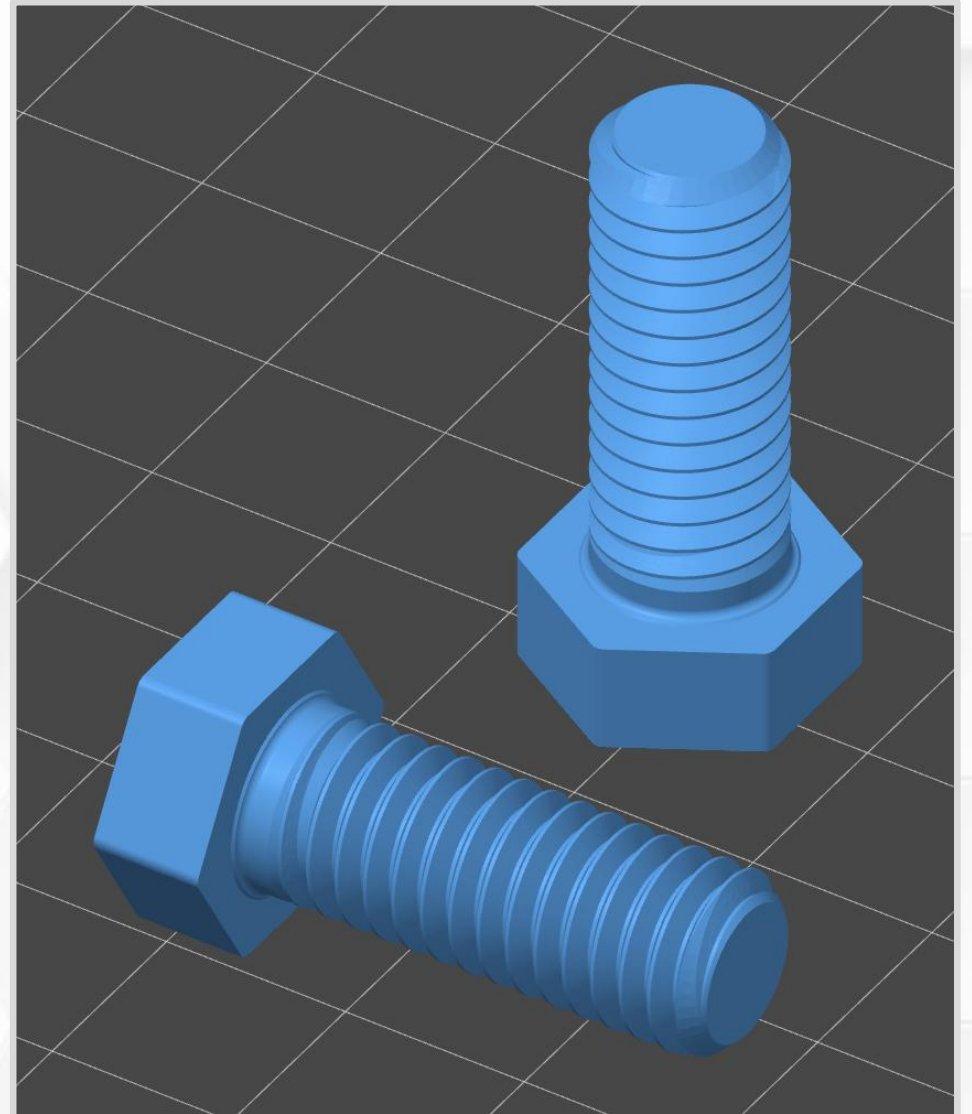
(Tips by
Billie Ruben)



Screws

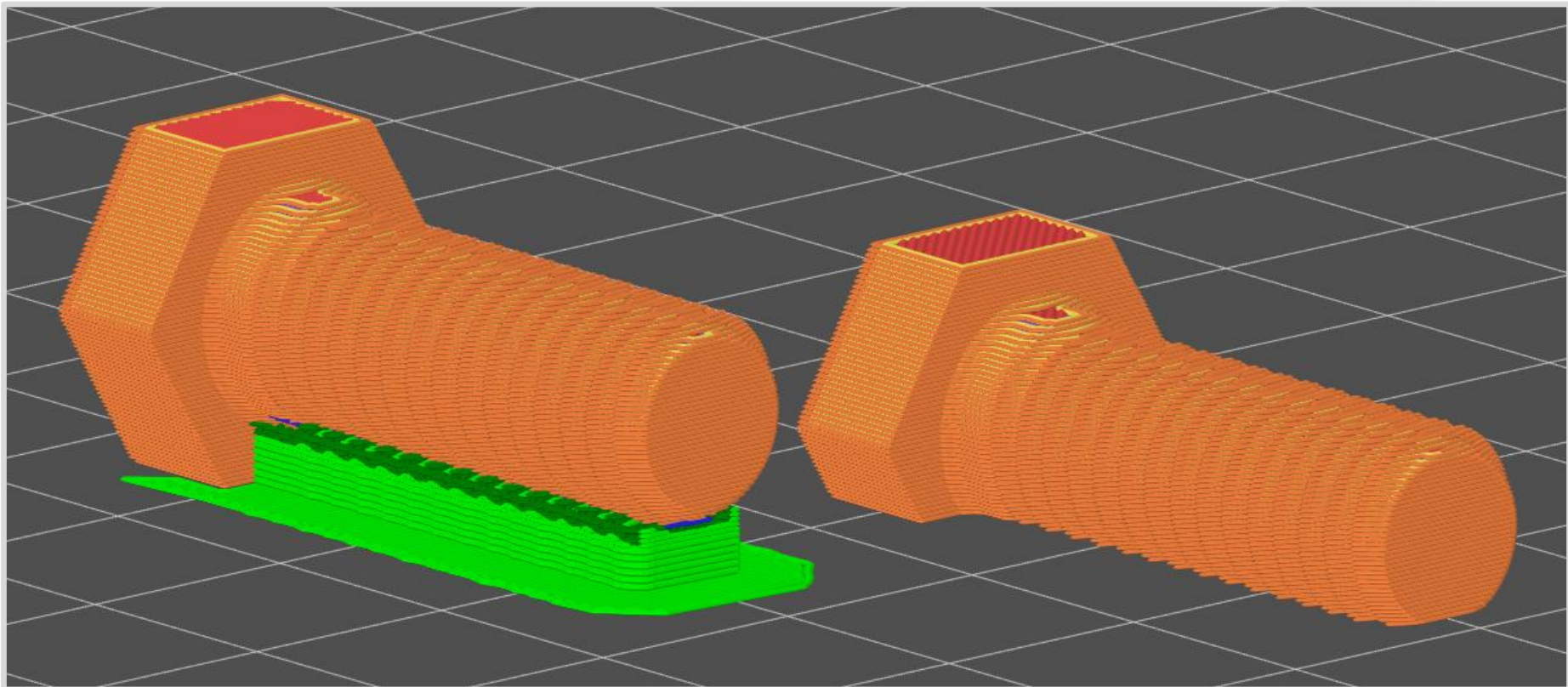
Let's 3D print a plastic screw.

- What differences can you see in **strength** and **printability**?
 - Compare and contrast print orientations.
 - Horizontal.
 - Layer lines maximize strength.
 - Nozzle must be smaller than threads.
 - Bottom threads are floating extrusions.
- Suggestions?



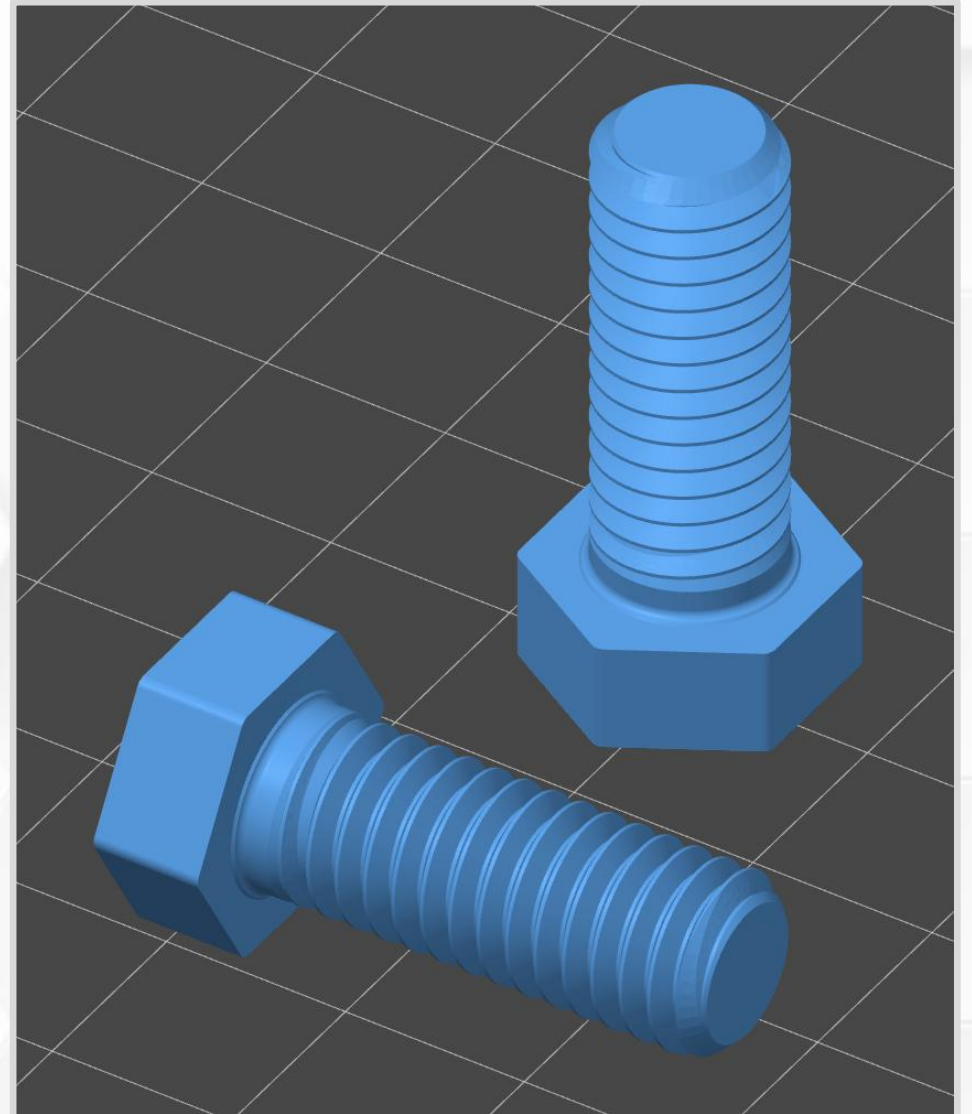
Screws

- Supports will probably leave a messy surface.
- Trim the bottom so it can sit flat. Clean threads and more bottom surface area.



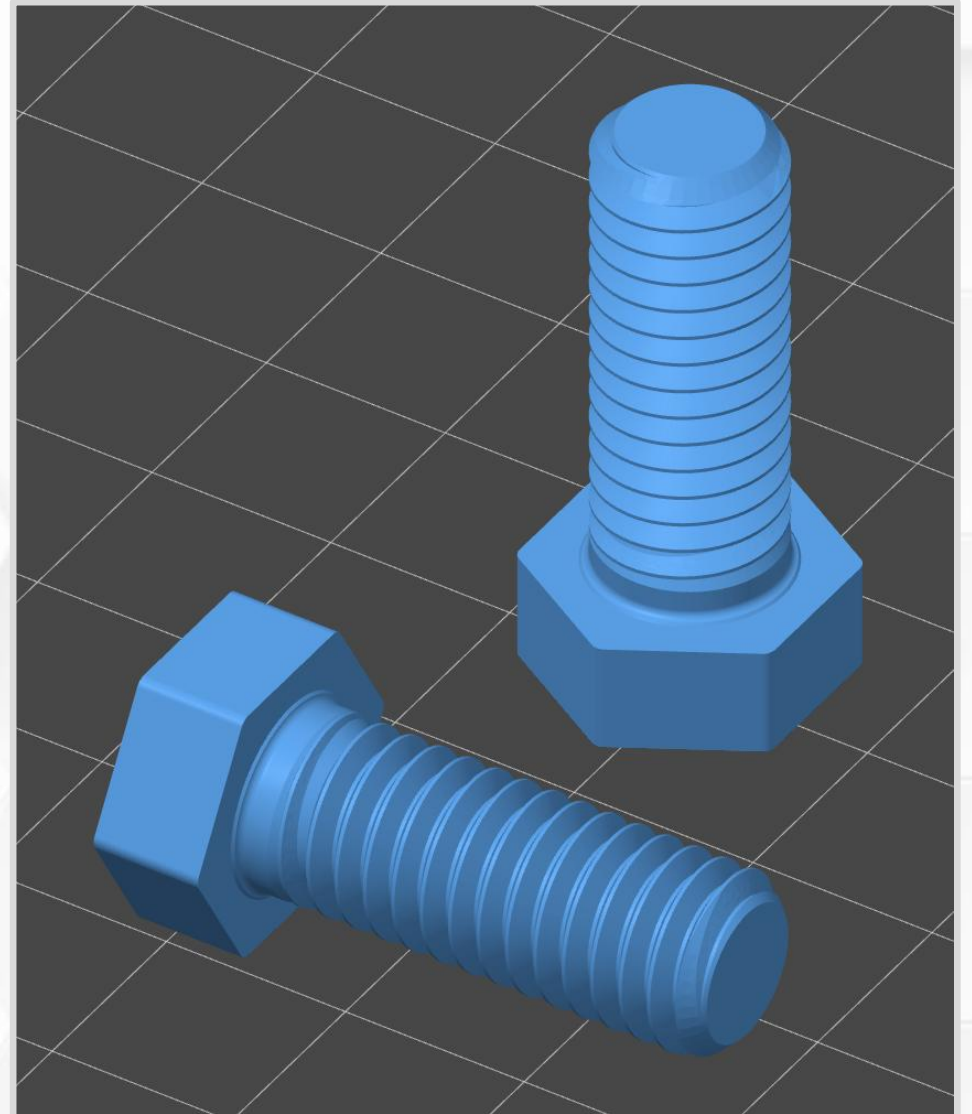
Screws

- Vertical
 - Thread size limited by layer height.
 - Typical threads create steep overhangs. Compensate with thinner layers.
 - Layers lines create a weaker part.
 - No supports.
 - No modifications required.



Further Reading

- Custom Threads for 3D printing by FlyingGyroscope.
 - Parametric template.
 - Includes CAD lesson.
- Recommended viewing.
 - Strength testing printed bolts by My Tech Fun.
 - And a follow-up video.



Snap Buckle

Let's design a replacement part.

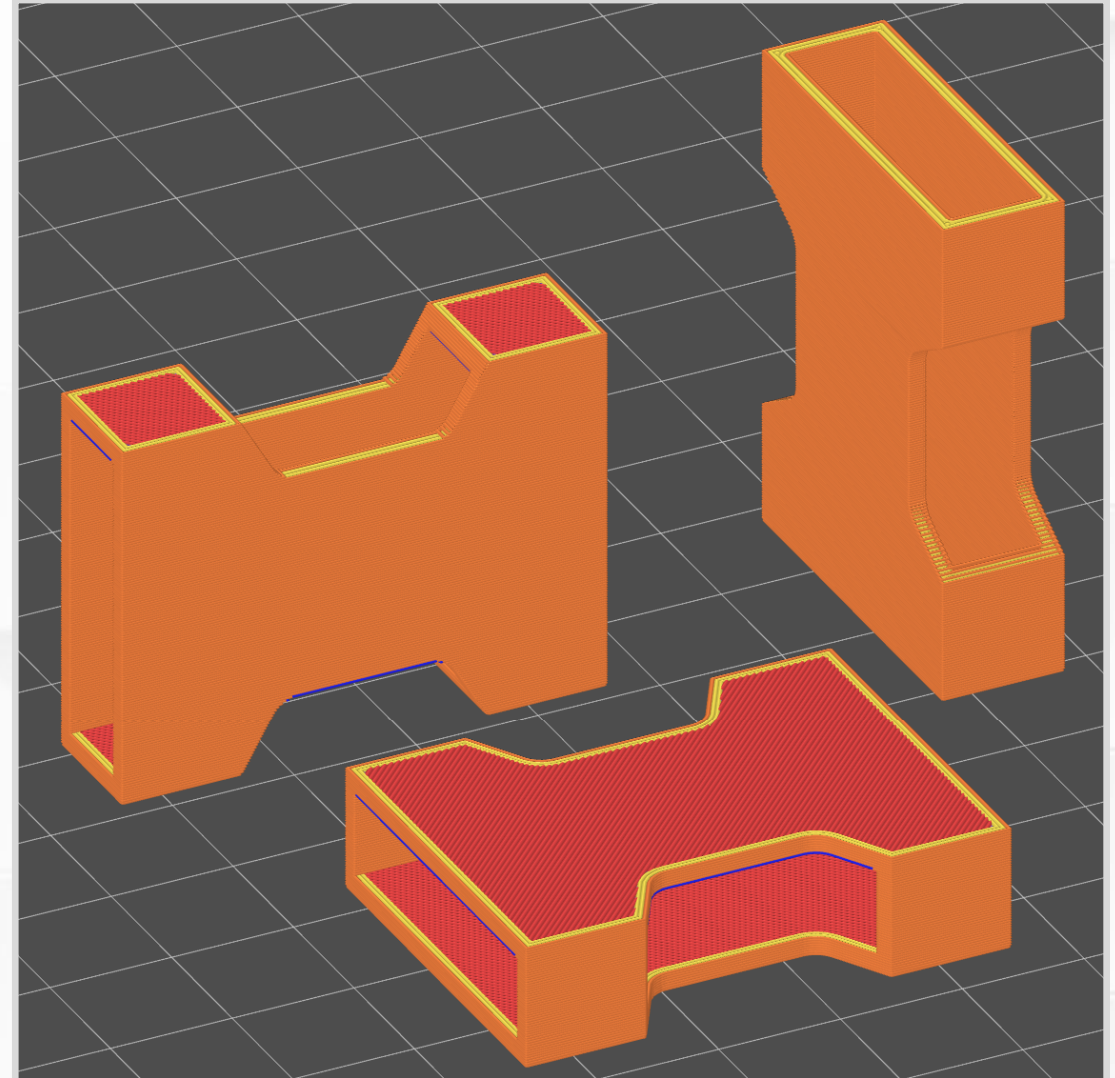
- Should you print a direct copy? Or does it need modifications?
- What 3D printing advantages and disadvantages do you see?



Snap Buckle

Let's compare several print orientations for the socket.

- How does each one balance **printability** and **part strength**?
- Can you modify the part for high printability and high strength?



Snap Buckle

- How does this design achieve high strength and high printability?



(Snap buckle by Squinn)

Snap Buckle

- **Horizontal orientation.**
- High strength for intended use.
- Perfect for the compliant mechanism.
- Good first layer contact.
- **The socket has been reimagined.**
- Dovetail (trapezoid) traps middle prong.
- Overhangs instead of bridges.

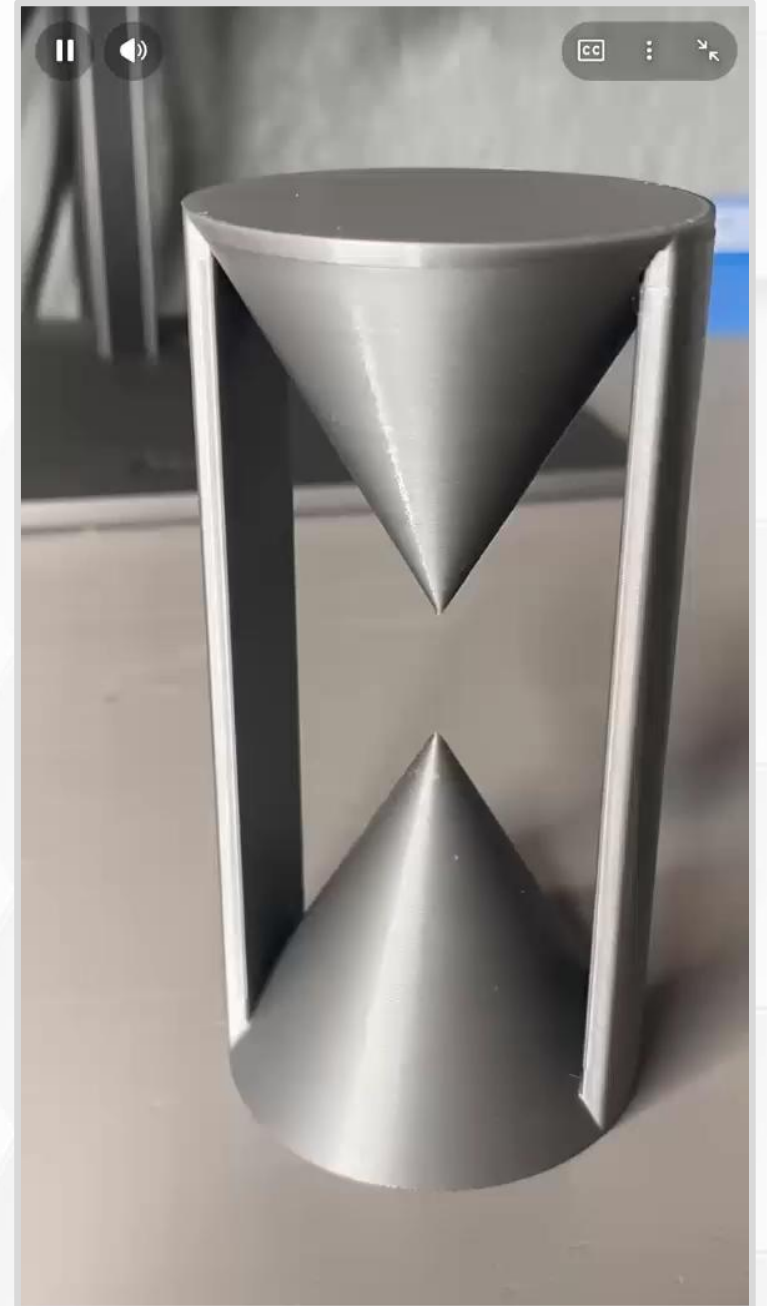
(Snap buckle by Squinn)



Creative Solutions

- I love the creativity of the maker community.
- This was printed without supports. How?
 - Put a printed part inside a printed part.

(Animation, YouTube short by christopherhelmke)





github.com/FlyingGyroscope/IntroTo3DPrinting/